Quiz 2 (Version 1)

CAS CS 132: *Geometric Algorithms*

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- ▷ You will have approximately 35 minutes to complete this quiz.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.

1 Spans and Linear Equations

Determine a linearly independent set of vectors V such that span V is the same as the solution set of the following linear equation.

$$2x - 3y + 4z = 0$$

Any non-co-linear pair of vectors that lie in the solution set form a valid solution

Solution.

$$V = \left\{ \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \\ 3 \end{bmatrix} \right\}$$

2 Dependence Relations

Let $A = [\mathbf{a}_1 \ \ \mathbf{a}_2 \ \ \mathbf{a}_3 \ \ \mathbf{a}_4]$ be a matrix (that is, $\mathbf{a}_1, \ldots, \mathbf{a}_4$ are the columns of A from left to right) that is row equivalent to the matrix given below. Write a dependence relation for the columns of A. That is, write the vector $\mathbf{0}$ as a nontrivial linear combination of the columns of A.

$$\begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 4 & -1 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -3 & 3 \\ 0 & 1 & 4 & -1 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -3 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$x_1 = 3x_3$

$x_2 = -4x_3$

x_3 is free

$x_4 = 0$

Solution.

$$3\vec{a}_1 - 4\vec{a}_2 + \vec{a}_3 = \vec{0}$$

3 Matrix Equations

Determine a general form solution for the homogeneous equation $A\mathbf{x} = \mathbf{0}$ where A is given below.

$$\begin{bmatrix} 1 & 3 & -5 \\ -2 & 2 & 2 \\ 3 & 2 & -8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & -5 \\ -2 & 2 & 2 \\ 3 & 2 & -8 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -5 \\ -1 & 1 & 1 \\ 3 & 2 & -8 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -5 \\ 0 & 4 & -4 \\ 0 & -7 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -5 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Solution.

$x_1 = 2x_3$

$x_2 = x_3$

x_3 is free

4 NumPy Array Slicing

Implement the function `quad` that, given a 2D NumPy array `A`, returns four 2D NumPy arrays corresponding to the four quadrants of `A`. For simplicity you may assume that `A` has an even number of rows and an even number of columns. In addition to array slicing you'll need to use the `shape` property and integer division (`//`). The following is an example of the expected behavior.

```
>>> A = np.array([[0, 1, 2, 3], [4, 5, 6, 7]])
>>> top_left, top_right, bottom_left, bottom_right = quad(A)
>>> top_left
array([[0, 1]])
>>> bottom_right
array([[6, 7]])
```

```
def quad(A):
    (m,n) = A.shape
    tl = A[0:m//2, 0:n//2]
    tr = A[0:m//2, n//2:]
    bl = A[m//2:, 0:n//2]
    br = A[m//2:, n//2:]
    return (tl, tr, bl, br)
```

Solution.